Symbolic Computation in Abel's Addition Theorem for Algebraic Integrals

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Addition Theorems on Integrals

Log case.

$$\log(u)+\log(v)=\log(uv) \quad \Leftrightarrow \quad \int_1^u \frac{1}{x}dx+\int_1^v \frac{1}{x}dx=\int_1^{uv} \frac{1}{x}dx$$

Proof.

$$\int_1^u \frac{1}{x}dx+\int_1^v \frac{1}{x}dx=\int_1^{T(u,v)} \frac{1}{x}dx \quad \Rightarrow \quad \frac{1}{u}=\frac{\partial T/\partial u}{T}$$

$$\Downarrow$$

$$T=C\cdot u, \quad \text{where } C \text{ is free of } u.$$

$$\text{Take } u=1, \text{ then } T=C=v. \text{ So } T=u\cdot v.$$

Addition Theorems on Integrals

Tangent case.

$$\tan(x+y) = \frac{\tan(x)+\tan(y)}{1-\tan(x)\tan(y)}$$

$$\Updownarrow$$

$$\arctan(u)+\arctan(v) = \arctan(\frac{u+v}{1-uv})$$

$$\Updownarrow$$

$$\int_0^u \frac{1}{1+x^2}dx + \int_0^v \frac{1}{1+x^2}dx = \int_0^{(u+v)/(1-uv)} \frac{1}{1+x^2}dx$$

Sine case.

$$\int_0^u \frac{1}{\sqrt{1-x^2}}dx + \int_0^v \frac{1}{\sqrt{1-x^2}}dx = \int_0^{u\sqrt{1-v^2}+v\sqrt{1-u^2}} \frac{1}{\sqrt{1-x^2}}dx$$

A Brief History

Elliptic Integral.

$$\int R(x, \sqrt{P(x)})dx,$$

where R is a rational function and $P(x)$ is a polynomial of degree 3 or 4 with rational coefficients and distinct roots.

Liouville. The elliptic integrals

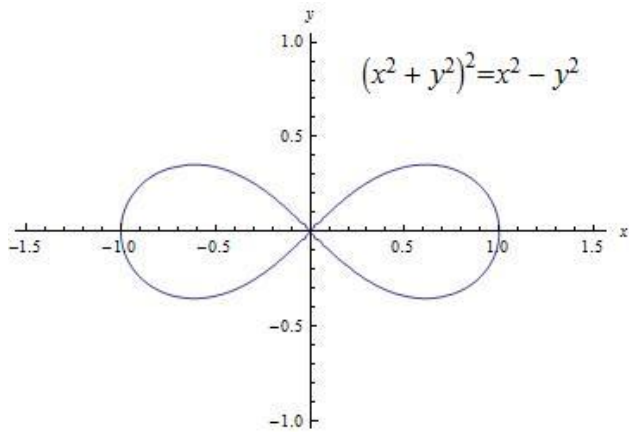
$$\int_0^u \frac{dx}{\sqrt{1-x^4}}, \quad \int_0^u \frac{1}{\sqrt{x(x-1)(x-2)}}dx, \ldots$$

are not elementary functions.

A Brief History

Fagnano considered the arc length of the lemniscate whose rectangular equation is $(x^2+y^2)^2=(x^2-y^2)$ and obtained that

$$\int_0^{T(u)} \frac{dx}{\sqrt{1-x^4}} = 2\int_0^{u} \frac{dy}{\sqrt{1-y^4}}, \qquad \text{where } T(u)=\frac{2u\sqrt{1-u^4}}{1+u^4}$$



A Brief History

Euler found the general solution of the differential equation:

$$\frac{dx}{\sqrt{1-x^4}} = \frac{dy}{\sqrt{1-y^4}} \Rightarrow x = \frac{y\sqrt{1-c^4}+c\sqrt{1-y^4}}{1+c^2y^2}.$$

This leads to the addition theorem:

$$\int_0^u \frac{dx}{\sqrt{1-x^4}} + \int_0^v \frac{dx}{\sqrt{1-x^4}} = \int_0^{T(u,v)} \frac{dx}{\sqrt{1-x^4}},$$

where

$$T(u,v) = \frac{u\sqrt{1-v^4}+v\sqrt{1-u^4}}{1+u^2v^2}.$$

A Brief History

Legendre.

$$\int R(x)/\sqrt{P(x)}dx$$

where $R(x)$ is a rational function and $\deg P(x) = 4$.
The elliptic integrals of the first, second and third kind:

$$\int \frac{dx}{\sqrt{1-x^2}\sqrt{1-k^2x^2}}, \int \frac{x^2dx}{\sqrt{1-x^2}\sqrt{1-k^2x^2}}, \int \frac{dx}{(x-a)\sqrt{1-x^2}\sqrt{1-k^2x^2}}.$$

Jacobi. The inversion of the elliptic integral $\rightarrow$ Elliptic function.

A Brief History

Weierstrass. Elliptic curve: $y^2 = 4x^3 - g_2x - g_3$.

$$f(u) = \int_0^u \frac{dx}{\sqrt{4x^3 - g_2x - g_3}},$$

where g_2 and g_3 are constants such that $g_2^3 - 27g_3^2 \neq 0$.

$$\int_0^{x_1} \frac{dx}{\sqrt{4x^3 - g_2x - g_3}} + \int_0^{x_2} \frac{dx}{\sqrt{4x^3 - g_2x - g_3}} = \int_0^{x_3} \frac{dx}{\sqrt{4x^3 - g_2x - g_3}},$$

where (x_3, y_3) is the intersection of the line through $(x_1, y_1), (x_2, y_2)$ and the elliptic curve.

A Brief History

Abel. Abelian integral:

$$\int R(x,y)dx,$$

where $R(x,y)$ is a rational function of x and y with the polynomial relation $F(x,y)=0$.
Note that abelian integral is the generalization of elliptic integral.

Abel's Theorem. G. A. Bliss: "In the history of mathematics there have been few instances in which as great an advance has been made by one writer at one time, and the theorem is one which every mathematician should know."

What is Abel's Theorem?

Theorem. Let C and C' be plane curves given by the equations

$$C: \ F(x,y) = 0,$$
$$C': \ G(x,y) = 0.$$

Note that the coefficients of $G(x,y)$ are rational functions of the coefficient variables $a_1, \ldots, a_k$. These curves have n points of intersection $(x_1,y_1), \ldots, (x_n,y_n)$. Consider the sum

$$I = \sum_{i=1}^{n} \int_{(x_0,y_0)}^{(x_i,y_i)} R(x,y)dx,$$

The sum I is equal to a rational function of the coefficient variables plus a linear combination of logarithms.

What is Abel's Theorem?

Addition Theorem. Let $n = 3$, then

$$I = \int_a^{x_1} R(x,y)dx + \int_a^{x_2} R(x,y)dx + \int_a^{x_3} R(x,y)dx$$

where a is the abscissa of a fixed point in the Riemann surface of $F(x,y) = 0$.

By some calculation an algebraic function T w.r.t. x_1 and x_2 can be obtained such that

$$\int_a^{x_1} R(x,y)dx + \int_a^{x_2} R(x,y)dx = \int_a^{T(x_1,x_2)} R(x,y)dx + \omega(x_1,x_2).$$

where $\omega(x_1,x_2)$ is an algebraic function or plus a logarithm term of an algebraic function.

Calculation of the Addition Formulae

Given an algebraic relation $F(x,y)=0$, firstly choose a proper curve $G(x,y)$, usually such that $n=3$.

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Symbolic-Computational Problem. Let $(x_1,y_1),\ldots,\ (x_n,y_n)$ be the points of intersection of the curves:

$$C:\ F(x,y)=0 \text{ and } C':\ G(x,y)=0.$$

How to compute the integral:

$$I=\sum_{i=1}^{n}\int_{a}^{x_i} R(x,y)dx.$$

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- Make an elimination on y from the curves C and C'(always compute a resultant of F and G w.r.t. y), then obtain that $\phi(x,a_1,\ldots,a_k)=0$ where $\deg_x(\phi)=n$.

Calculation of the Addition Formulae

- By Vieta's Theorem, elementary symmetric polynomials $\sigma_1, \ldots, \sigma_n$ of $x_1, \ldots, x_n$ can be written as rational functions of $a_1, \ldots, a_k$.

$$\sigma_i = r_i(a_1, \ldots, a_k), \text{ for each } i = 1, \ldots, n.$$

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- Take $\partial / \partial a_j$ on both sides of $\phi = 0$ we have that x_{a_j} is a rational function of $x, a_1, \ldots, a_k$ for each $j = 1, \ldots, k$.

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- Take $\partial/\partial a_j$ on both sides of $\phi = 0$ we have that x_{a_j} is a rational function of $x, a_1, \ldots, a_k$ for each $j = 1, \ldots, k$.
- Assume that y can be written as a polynomial of x, maybe related to $a_1, \ldots, a_k$.

Calculation of the Addition Formulae

- Compute $dI = \sum_{j=1}^{k} \sum_{i=1}^{n} R(x_i, y_i)(\partial x_i / \partial a_j) da_j$.
  Let

$$f_j = \sum_{i=1}^{n} R(x_i, y_i)(\partial x_i / \partial a_j).$$

  Then f_j is a symmetric in $x_1, \ldots, x_n$.

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The Fundamental Theorem of Symmetric Polynomials.

$\downarrow$

Simplify f_j as a rational function of $a_1, \ldots, a_k$.

Calculation of the Addition Formulae

- Compute

$$I = \sum_{i=1}^{n} \int_{a}^{x_i} R(x,y)dx.$$

Subproblem 2. Given $dI = \sum_{j=1}^{k} f_j \, da_j$ where f_j is a rational function of $a_1, \ldots, a_k$ for each j, how to compute the integral I ?

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Symbolic Integration

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I is a rational function plus a linear combination of logarithms.

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Symbolic Integration

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I is a rational function plus a linear combination of logarithms.

- Through Vieta's relation compute x_3 and $a_1, \ldots, a_k$ as an algebraic function of x_1 and x_2, then transform the equation into the addition theorem.

Addition Theorem on Elliptic Integrals

Let

$$E(k,z)=\int_0^z \frac{dx}{\sqrt{(1-x^2)(1-k^2x^2)}},$$

compute $T(x_1,x_2)$ s.t. $E(k,x_1)+E(k,x_2)=E(k,T(x_1,x_2))$.

- Choose the curves C and C':

$$\begin{cases} y=\sqrt{(1-x^2)(1-k^2x^2)} \\ y=kx^2+ax+b \end{cases}$$

$$\Rightarrow \quad \phi(a,b,x)=2akx^3+(a^2+k^2+2bk+1)x^2+2abx+b^2-1=0.$$

Addition Theorem on Elliptic Integrals

- $$\begin{cases} \phi_x \cdot x_a + \phi_a = 0 & \Rightarrow x_a = -\dfrac{x(kx^2+ax+b)}{3akx^2+(a^2+k^2+2bk+1)x+ab} \\ \phi_x \cdot x_b + \phi_b = 0 & \Rightarrow x_b = -\dfrac{kx^2+ax+b}{3akx^2+(a^2+k^2+2bk+1)x+ab} \end{cases}$$

- By Vieta's formulae, $$\begin{cases} x_1+x_2+x_3 = -\dfrac{a^2+k^2+2bk+1}{2ak} \\ x_1x_2+x_1x_3+x_2x_3 = \dfrac{b}{k} \\ x_1x_2x_3 = \dfrac{1-b^2}{2ak} \end{cases}.$$

Substitute y_i by $kx_i^2+ax_i+b$ in $R(x,y)=1/y$.

Through the algorithm executed on Maple we can simplify the expressions and figure out $dI=0$, i.e. $I=C$.

Addition Theorem on Elliptic Integrals

- Fix $\bar{x}_1 = 0$ and $\bar{x}_3 = x_3$, we have $\bar{b} = 1$ and $\bar{x}_2, \bar{a}$ are flexible.
  Since $E(k,0) = 0$,

$$\begin{cases} E(k,x_1)+E(k,x_2)+E(k,x_3) = C \\ E(k,\bar{x}_2)+E(k,x_3) = C \end{cases} \Rightarrow E(k,x_1)+E(k,x_2) = E(k,\bar{x}_2).$$

By Vieta's formulae, $\begin{cases} \bar{x}_2 + x_3 = -\dfrac{\bar{a}^2+k^2+2k+1}{2\bar{a}k} \\ \bar{x}_2 x_3 = \dfrac{1}{k} \end{cases}$.

Then $\bar{x}_2 = \dfrac{1}{kx_3}$, which implies that

$$E(k,x_1)+E(k,x_2) = E(k,\frac{1}{kx_3}).$$

Addition Theorem on Elliptic Integrals

- By $\begin{cases} y_1 = kx_1^2 + ax_1 + b \\ y_2 = kx_2^2 + ax_2 + b \end{cases}$ and $\begin{cases} y_1 = \sqrt{(1-x_1^2)(1-k^2x_1^2)} \\ y_2 = \sqrt{(1-x_2^2)(1-k^2x_2^2)} \end{cases}$,

$$a = \frac{y_1 - y_2 - k(x_1^2 - x_2^2)}{x_1 - x_2}$$
$$= \frac{\sqrt{(1-x_1^2)(1-k^2x_1^2)} - \sqrt{(1-x_2^2)(1-k^2x_2^2)} - k(x_1^2 - x_2^2)}{x_1 - x_2},$$
$$b = y_1 - kx_1^2 - ax_1 = \sqrt{(1-x_2^2)(1-k^2x_2^2)} - kx_1^2 - ax_1.$$

Since $x_3 = \dfrac{b/k - x_1x_2}{x_1 + x_2}$, we can obtain the algebraic function T:

$$T(x_1, x_2) = \bar{x}_2 = \frac{1}{kx_3} = \frac{x_1\sqrt{(1-x_2^2)(1-k^2x_2^2)} + x_2\sqrt{(1-x_1^2)(1-k^2x_1^2)}}{1 - k^2x_1^2x_2^2}.$$

Thank You!